

CAR-bench: Evaluating the Consistency and Limit-Awareness of LLM Agents under Real-World Uncertainty

CAR-bench: 在现实世界不确定性下评估大语言模型代理的一致性与限制意识

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Abstract

摘要

Existing benchmarks for Large Language Model (LLM) agents focus on task completion under idealistic settings but overlook reliability in real-world, user-facing applications. In domains, such as in-car voice assistants, users often issue incomplete or ambiguous requests, creating intrinsic uncertainty that agents must manage through dialogue, tool use, and policy adherence. We introduce CAR-bench, a benchmark for evaluating consistency, uncertainty handling, and capability awareness in multi-turn, tool-using LLM agents in an in-car assistant domain. The environment features an LLM-simulated user, domain policies, and 58 interconnected tools spanning navigation, productivity, charging, and vehicle control. Beyond standard task completion, CAR-bench introduces Hallucination tasks that test agents' limit-awareness under missing tools or information, and Disambiguation tasks that require resolving uncertainty through clarification or internal information gathering. Baseline results reveal large gaps between occasional and consistent success on all task types. Even frontier reasoning LLMs achieve less than 50% consistent pass rate on Disambiguation tasks due to premature actions, and frequently violate policies or fabricate information to satisfy user requests in Hallucination tasks, underscoring the need for more reliable and self-aware LLM agents in real-world settings.¹

现有大型语言模型（LLM）代理基准测试主要关注在理想化环境下任务完成的情况，但忽略了在现实世界、面向用户的实际应用中的可靠性。在诸如车载语音助手等领域中，用户经常发出不完整或模糊的请求，这会带来固有的不确定性，代理必须通过对话、工具使用和政策遵循来管理这些不确定性。我们引入了 CAR-bench，这是一个用于评估多轮对话、使用工具的 LLM 代理在车载助手领域中的一致性、不确定性处理和意识能力的基准测试。该环境包含一个由 LLM 模拟的用户、领域政策以及涵盖导航、生产力、充电和车辆控制的 58 个相互关联的工具。除了标准的任务完成之外，CAR-bench 还引入了幻觉任务，用于测试代理在缺少工具或信息时的边界意识，以及歧义消除任务，要求通过澄清或内部信息收集来解决不确定性。基准测试结果揭示了在所有任务类型中偶然成功与持续成功之间的巨大差距。即使是前沿推理的 LLM，在歧义消除任务中也由于过早行动而实现不到 50% 的持续通过率，并且在幻觉任务中经常违反政策或编造信息以满足用户请求，这突显了在现实世界环境中需要更加可靠和自我意识的 LLM 代理的必要性。¹

1 Introduction

1 引言

LLM agents are transforming human-computer interaction, moving beyond single-turn question-answering toward autonomous execution of complex, multi-step tasks (Acharya et al., 2025; Wang et al., 2024; Hosseini and Seilani, 2025). Deploying such agents, however, requires more than potential capability; it demands consistent performance

LLM agents 正在改变人机交互方式，从单轮问答模式转向自主执行复杂、多步骤任务（Acharya 等，2025；Wang 等，2024；Hosseini 和 Seilani，2025）。然而，部署此类代理不仅需要潜在的能力，还要求其具备一致的性能表现。

¹ Code is released at <https://github.com/CAR-bench/car-bench>.

¹ 代码已发布在 <https://github.com/CAR-bench/car-bench>。

and a calibrated awareness of their own limitations across multi-turn interactions.

并且在多轮交互中具备对自身局限性的校准意识。

Existing benchmarks evaluate agents in idealized settings: Tool-use benchmarks (Huang et al., 2024b; Guo et al., 2024) isolate API-calling capabilities without considering conversation; TravelPlanner (Xie et al., 2024) and ToolLLM (Qin et al., 2024) evaluate single-turn interactions with

complete task information provided upfront, while BFCLv3 (Patil et al., 2025) and ToolTalk (Farn and Shin, 2023) rely on pre-collected, off-policy trajectories, conditioning agents on idealized histories. τ -bench (Yao et al., 2024) advanced this line of work by introducing three key requirements for real-world agents: (1) dynamic interaction with both humans and programmatic APIs, (2) adherence to domain-specific rules and policies, and (3) consistency and robustness across repeated trials. By combining a simulated user, API tools, and policy constraints, τ -bench provided the first dynamic and policy-guided benchmark, while ToolSandbox (Lu et al., 2025) extended this to state-dependent tool interactions.

现有的基准测试在理想化的环境中评估智能体：工具使用基准（Huang 等人，2024b；Guo 等人，2024）隔离了 API 调用能力，而不考虑对话；TravelPlanner (Xie 等人，2024) 和 ToolLLM (Qin 等人，2024) 评估单轮交互，且在一开始就提供完整的任务信息，而 BFCLv3 (Patil 等人，2025) 和 ToolTalk (Farn 和 Shin, 2023) 则依赖于预先收集的、离策略轨迹，将智能体条件化在理想化的历史记录上。 τ -bench (Yao 等人，2024) 通过引入三个关键要求，推动了这一研究方向的发展，适用于现实中的智能体：(1) 与人类和程序化 API 动态交互，(2) 遵守特定领域的规则和政策，(3) 在重复试验中保持一致性和鲁棒性。通过结合模拟用户、API 工具和策略约束， τ -bench 提供了首个动态且受策略引导的基准测试，而 ToolSandbox (Lu 等人，2025) 则将其扩展到状态依赖的工具交互。

However, two critical deployment challenges remain unaddressed. First, many user requests are unsatisfiable: required tools might be missing, tool parameters lack sufficient granularity, or environment queries return incomplete data. Recent work highlights that LLMs are rewarded for producing plausible completions rather than admitting uncertainty (Kalai et al., 2025), making such cases especially prone to hallucinations. While Tool-Sandbox and BFCLv3 consider missing functions, they overlook these broader failure modes. Second, agents face ambiguity from underspecified user requests or incomplete observation. Resolving ambiguity requires meta-reasoning: deciding which actions or clarifications maximize information gain (Kobalczyk et al., 2025). Real-world deployment demands a paradigm shift: from evaluating only correct tool execution to assessing whether agents

然而，仍然存在两个关键的部署挑战未被解决。首先，许多用户请求是无法满足的：所需的工具可能缺失，工具参数缺乏足够的精细度，或者环境查询返回的数据不完整。近期的研究指出，大型语言模型（LLMs）会因生成合理的补全而受到奖励，而非承认不确定性（Kalai 等，2025），使得这些情况特别容易产生幻觉。虽然 Tool-Sandbox 和 BFCLv3 考虑了缺失的功能，但它们忽略了这些更广泛的失败模式。其次，代理会因用户请求描述不明确或观察信息不完整而面临模糊性。解决模糊性需要元推理：决定哪些行动或澄清能够最大化信息增益（Kobalczyk 等，2025）。现实世界的部署需要一种范式转变：从仅评估正确的工具执行，转变为评估代理是否能够.....

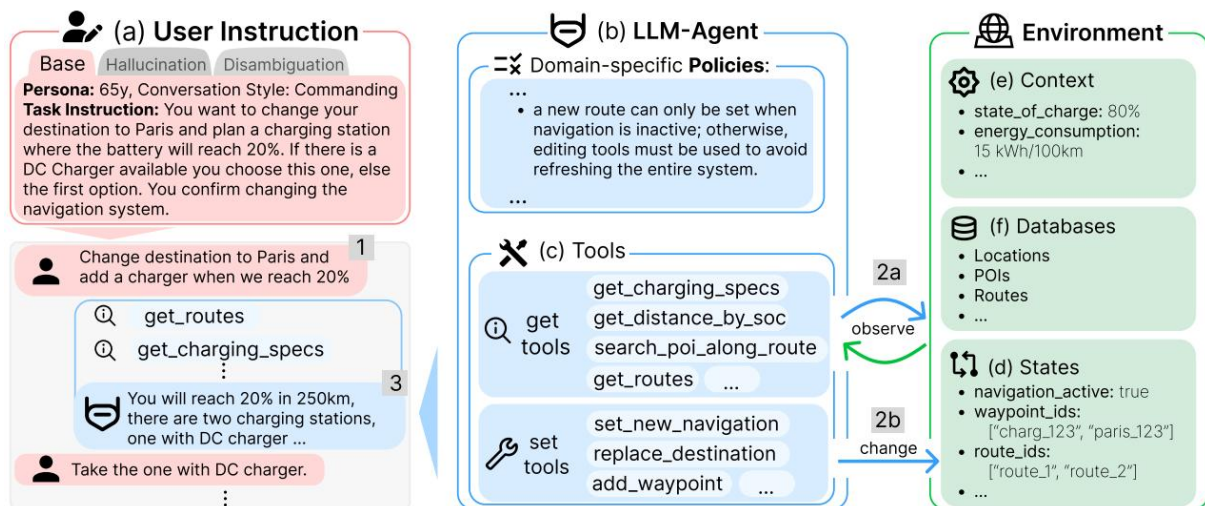


Figure 1: images/3d3d458fddb3aa1fe5786272c7e3bcc35b5a8fb6197e5d54c00655a8771132a0.jpg

Figure 1: Overview of the CAR-bench components. (a) An LLM-simulated user generates multi-turn messages following task instructions (1); (b) the LLM agent, guided by domain policies, interacts with (c) tools to (2a) observe the environment or (2b) modify its state, until producing an informed response (3). The environment consists of (d) mutable states, (e) fixed context variables, and (f) static databases. The user instructions show the Base task type, the task types Hallucination and Disambiguation are explained in Section 3.2.

图 1: CAR-bench 组件概述。(a) 一个 LLM 模拟的用户根据任务指令生成多轮对话消息 (1); (b) 在领域策略指导下, LLM 代理与 (c) 工具进行交互, 以 (2a) 观察环境或 (2b) 修改其状态, 直到生成一个有根据的回复 (3)。环境由 (d) 可变状态、(e) 固定上下文变量和 (f) 静态数据库组成。用户指令显示了基础任务类型, 任务类型“幻觉”和“消歧”在第 3.2 节中进行了解释。

reliably recognize when they cannot do this or cannot safely do this yet rather than act anyway.

可靠地识别他们无法做到这一点或暂时无法安全地做到这一点的情况, 而不是盲目行动。

The automotive in-car assistant domain serves as a natural testbed for this paradigm: ambiguous, speech-like requests from non-expert users, heterogeneous vehicle-specific APIs, strict safety constraints, and driver distraction limitations (Strayer et al., 2016) that make hallucination avoidance and correct disambiguation safety-critical.

车载助手领域是这一范式的一个自然测试平台: 非专业用户提出的模糊、类似语音的请求, 车辆特定的异构 API, 严格的安全部署约束以及驾驶员注意力限制 (Strayer 等, 2016), 这些因素使得避免幻觉和正确消歧成为安全至关重要的问题。

We introduce CAR-bench, the first benchmark to systematically evaluate consistency, uncertainty handling, and capability awareness in multi-turn, policy-constrained LLM agents, using the car-domain as a uniquely demanding testbed. CAR-bench comprises six components (see Figure 1): (a) an LLM-simulated user, following task instructions, (b) an agent guided by 19 domain policies, (c) a comprehensive agent toolset for information retrieval and action execution, and an interactive environment with (d) mutable states, (e) fixed context variables, and (f) contextual databases. Beyond standard task completion, CAR-bench introduces two new task types: (1) Hallucination tasks, testing whether agents acknowledge missing capabilities or data rather than fabricating, and (2) Disambiguation tasks, evaluating whether agents resolve uncertainty before taking actions, either through internal information gathering or user clarification.

我们介绍了 CAR-bench, 这是首个用于系统评估多轮、受策略约束的 LLM 代理中一致性、不确定性处理和能力意识的基准, 使用 cardomain 作为独特且具有挑战性的测试平台。CAR-bench 包含六个组件 (参见图 1): (a) 一个遵循任务指令的 LLM 模拟用户, (b) 由 19 个领域策略引导的代理, (c) 用于信息检索和动作执行的全面代理工具集, 以及一个具有 (d) 可变状态、(e) 固定上下文变量和 (f) 上下文数据库的交互环境。除了标准任务完成之外, CAR-bench 还引入了两种新的任务类型: (1) 幻觉任务, 测试代理是否承认缺失的能力或数据, 而不是编造, 以及 (2) 消歧任务, 评估代理是否在采取行动之前解决不确定性, 无论是通过内部信息收集还是用户澄清。

To measure deployment readiness, we report Pass³ (success in all $k = 3$ trials) alongside Pass@3 (at least one success). Baseline evalua-

为了衡量部署准备情况, 我们报告 Pass³ (在所有 $k = 3$ 次试验中均成功) 和 Pass@3 (至少有一次成功)。基线评估-

	Conversational	State-Depend.	Halluci-nation	Disambiguation
CAR-bench	□	□	□	□
τ-bench	□	□	□	□
ToolSandbox	□	□	□	□
BFCLv3	□	□	□	□

	对话式	状态相关。	幻觉	释义
CAR-bench	□	□	□	□
τ -bench	□	□	□	□
工具沙盒	□	□	□	□
BFCLv3	□	□	□	□

Table 1: Comparison of related benchmarks. ToolSandbox and BFCLv3 evaluate missing tools but omit missing parameters and environment observations (partial Hallucination coverage).

表 1: 相关基准的比较。ToolSandbox 和 BFCLv3 评估了缺失的工具, 但忽略了缺失的参数和环境观察(部分幻觉覆盖)。

tion reveals that even SOTA models achieve only 54% average Pass³, with substantial consistency gaps: on Disambiguation alone, GPT-5 drops from 68% Pass@3 to 36% Pass³. Thinking models outperform non-thinking variants in task performance, with the gap widening as task complexity increases. Both novel task types prove challenging: Hallucination tasks expose non-thinking models’ tendency to fabricate rather than acknowledge limitations, with thinking models improving but plateauing at 60% Pass³; Disambiguation remains hardest, with no model exceeding 50% Pass³. Error analysis reveals a systematic completion-compliance tension: models prioritize satisfying user requests over following policies, leading to premature actions without complete information, stochastic policy violations, and fabricated responses when capabilities are missing.

结果表明,即使是最先进的模型(SOTA models)也只能达到 54% 的平均 Pass³, 存在显著的一致性差距: 仅在消歧任务 (Disambiguation) 上思考型模型在任务表现上优于非思考型模型, 随着任务复杂性的增加, 这种差距进一步扩大。两种新型任务类型都具有挑战性: 幻觉任务(Hallucination tasks)暴露出非思考型模型倾向于编造而非承认自身局限性, 思考型模型虽然有所提升,但最终在 Pass³ 上达到 60% 后趋于平稳; 消歧任务 (Disambiguation) 仍然是最难的任务, 没有任何模型的 Pass³ 超过 50%。错误分析揭示了一种系统性的完成与合规之间的张力: 模型优先满足用户请求而非遵循政策, 导致在信息不完整时采取过早行动, 随机违反政策, 以及在能力缺失时编造回应。

Contributions. (1) We provide a comprehensive evaluation framework with 58 interconnected tools and 19 domain policies in an automotive assistant environment. (2) We introduce Hallucination and Disambiguation task types to systematically assess limit-awareness and uncertainty resolution. (3) We present a error taxonomy across task types and a comparative analysis of reasoning versus non-reasoning models.

贡献。(1) 我们提供了一个全面的评估框架, 在汽车助手环境中包含 58 个相互关联的工具和 19 个领域政策。(2) 我们引入了“幻觉”和“消歧”任务类型, 以系统地评估对限制的意识 and 不确定性解决能力。(3) 我们展示了跨任务类型的错误分类, 并对推理模型与非推理模型进行了比较分析。

2 Related Work

2 相关工作

LLM Agent Benchmarks. Recent benchmarks evaluate LLM agents across diverse environments such as web navigation (Yao et al., 2022; Zhou et al., 2024), embodied tasks (Shridhar et al., 2021), or code execution (Jimenez et al., 2024). Within this landscape, tool-use benchmarks assess agents’ ability to perform single, parallel, and sequential API calls (Patil et al., 2025; Qin et al., 2024; Huang et al., 2024b; Li et al., 2023; Huang et al., 2024a), however, typically assuming full tool coverage and complete task information. More recent efforts such as BFCLv3 (Patil et al., 2025) and ToolSandbox (Lu et al., 2025) begin exploring capability awareness by withholding certain tools, but they still overlook subtler real-world limitations such as missing parameters or incomplete outputs. While most benchmarks prioritize singular task completion, τ -bench (Yao

et al., 2024) introduces consistency evaluation through its Pass^k metric, measuring reliability across repeated trials.

LLM Agent Benchmarks. 最近的基准测试评估了 LLM 代理在各种环境中的表现，例如网页导航 (Yao 等, 2022; Zhou 等, 2024)、具身任务 (Shridhar 等, 2021) 或代码执行 (Jimenez 等, 2024)。在这一领域中，工具使用基准测试评估代理执行单个、并行和顺序 API 调用的能力 (Patil 等, 2025; Qin 等, 2024; Huang 等, 2024b; Li 等, 2023; Huang 等, 2024a)，然而，通常假设工具覆盖完全且任务信息完整。近期的一些努力，如 BFCLv3 (Patil 等, 2025) 和 ToolSandbox (Lu 等, 2025)，开始通过隐藏某些工具来探索能力意识，但它们仍然忽略了更细微的现实限制，例如缺少参数或输出不完整。尽管大多数基准测试优先考虑单一任务完成，但 τ -bench (Yao 等, 2024) 通过其 Pass^k 指标引入了一致性评估，衡量在重复试验中的可靠性。

Multi-turn Dialogue and User Simulation. Real deployments require agents to manage iterative, under-specified user interactions rather than single-turn tasks. Existing multi-turn benchmarks often rely on pre-collected trajectories (Patil et al., 2025; Farn and Shin, 2023; Chen et al., 2021; Budzianowski et al., 2018), which, however, condition models on continuing idealized interaction histories rather than testing their policy development. To address this, recent work employs LLM-based user simulators that generate dynamic, on-policy conversations (Mu et al., 2024; Sekulic et al., 2024; Kong et al., 2024; Yao et al., 2024; Lu et al., 2025). These simulators, guided by user personas or scripts, capture realistic conversational variability (Kirmayr et al., 2025; Park et al., 2023). However, most work focuses on information exchange improvement rather than evaluating how agents detect and resolve incomplete or ambiguous requests.

多轮对话和用户模拟。实际部署要求代理管理迭代的、未明确指定的用户交互，而不仅仅是单轮任务。现有的多轮基准测试通常依赖于预先收集的轨迹 (Patil 等, 2025; Farn 和 Shin, 2023; Chen 等, 2021; Budzianowski 等, 2018)，然而，这些方法却使模型基于继续理想化的交互历史进行条件建模，而不是测试其策略开发能力。为了解决这一问题，近期的工作采用基于大语言模型 (LLM) 的用户模拟器，生成动态的、符合策略的对话 (Mu 等, 2024; Sekulic 等, 2024; Kong 等, 2024; Yao 等, 2024; Lu 等, 2025)。这些模拟器在用户角色或脚本的指导下，捕捉到现实对话中的变化性 (Kirmayr 等, 2025; Park 等, 2023)。然而，大多数工作关注的是信息交换的改进，而不是评估代理如何检测 and 解决不完整或模糊的请求。

Hallucination, Disambiguation, and Reasoning. LLMs frequently hallucinate plausible yet incorrect information instead of acknowledging uncertainty, a behavior reinforced by training objectives that reward completion (Kalai et al., 2025; Huang et al., 2025). Disambiguating unclear instructions, in contrast, requires meta-cognitive reasoning to identify missing information and determine clarification strategies (Kobalczyk et al., 2025). Reasoning-enhanced LLMs such as GPT-5 (OpenAI, 2025), are expected to improve on these aspects by reflective decision-making. Yet, assessing whether such reasoning actually mitigates these factors, and how consistently, remains underexplored.

幻觉、消歧义和推理。大语言模型 (LLMs) 经常在不确定的情况下生成看似合理但实际上错误的信息，而不是承认不确定性，这种行为受到训练目标的强化，这些目标奖励完成任务 (Kalai 等, 2025; Huang 等, 2025)。相比之下，澄清模糊的指令需要元认知推理，以识别缺失的信息并确定澄清策略 (Kobalczyk 等, 2025)。增强推理能力的大型语言模型，例如 GPT-5 (OpenAI, 2025)，预计通过反思性决策来改进这些方面。然而，评估这种推理是否确实缓解了这些因素，以及其一致性如何，仍然缺乏充分的研究。

Domain-Specific Policies. Real-world deployments require adherence to domain-specific policies beyond general world knowledge. τ -bench and τ^2 -bench (Yao et al., 2024; Barres et al., 2025) introduced policy-guided agents for retail, airline, and telecom domains. Policy adherence is especially relevant in the car-domain, where safety-critical vehicle APIs (Stappen et al., 2023) make hallucinations dangerous and managing driver distraction (Strayer et al., 2016) requires efficient disambiguation.

领域特定政策。现实部署需要遵循超出一般世界知识的领域特定政策。 τ -bench 和 τ^2 -bench (Yao 等人, 2024; Barres 等人, 2025) 为零售、航空和电信领域引入了基于政策的智能体。在汽车领域，遵循政策尤为重要，因为在该领域中，安全关键型车辆 API (Stappen 等人, 2023) 使得幻觉具有危险性，而管理驾驶员分心 (Strayer 等人, 2016) 则需要高效的歧义消除。

3 CAR-bench

3 CAR-bench

CAR-bench establishes a dynamic benchmark in the automotive domain by simulating an in-car assistant environment.

CAR-bench 通过模拟车内助手环境，建立了汽车领域的一个动态基准。

3.1 Benchmark Components

3.1 基准组件

The benchmark components and their specific instances are grounded in domain expertise and aim to reflect the complexity of real-world in-car assistant environments (ref. Figure 1).

基准组件及其具体实例基于领域专业知识，并旨在反映现实世界车载助手环境的复杂性（参见图 1）。

LLM-simulated User. The user is simulated by an LLM following detailed instructions for a specific multi-turn interaction. Each user is assigned a persona and task instruction. The persona includes three attributes: age (18–65), conversation style, and technical proficiency. Conversation styles - commanding, conversational, and questioning - are based on empirical in-car interaction insights. Technical proficiency is categorized as familiar with automotive terminology, regular (preferring everyday terms), or unspecified (cf. (Barres et al., 2025)). Task instructions define goals, information disclosure order, reaction rules, and completion criteria. In each turn, the user outputs a text message (visible to the agent) and a control word (meta-level

LLM 模拟用户。该用户由一个遵循详细指令进行特定多轮交互的 LLM 模拟而成。每个用户都会被分配一个角色和任务指令。角色包括三个属性：年龄（18–65 岁）、对话风格和专业技术水平。对话风格包括命令型、对话型和提问型，这些风格基于对车内交互的实证研究。专业技术水平分为熟悉汽车术语、普通（偏好日常用语）或未指定（参见（Barres 等，2025））。任务指令定义了目标、信息披露顺序、反应规则和完成标准。在每一轮中，用户输出一条文本消息（对代理可见）和一个控制词（元级别）

annotation). Control words enable (1) dynamic termination of variable-length interactions and (2) automated correctness evaluation. Extending τ -bench (Yao et al., 2024), Base tasks use ‘continue’ (proceed), ‘stop’ (goal reached), or ‘out of scope’ (agent diverged beyond user instructions) as control words; extensions for Hallucination and Disambiguation tasks are introduced in Section 3.2. User and agent interact exclusively via text messages: the user cannot observe tool calls or environment data; the agent cannot see control words.

注释)。控制词能够实现（1）对可变长度交互的动态终止和（2）自动正确性评估。扩展 τ -bench（Yao 等人，2024），基础任务使用“continue”（继续）、“stop”（目标达成）或“out of scope”（代理偏离了用户指令）作为控制词；关于幻觉和消歧任务的扩展内容将在第 3.2 节中介绍。用户和代理仅通过文本消息进行交互：用户无法观察到工具调用或环境数据；代理也无法看到控制词。

LLM-Agent with domain-specific policies. Our benchmark employs agents based on LLMs with native tool-calling support for baseline testing, shown by Yao et al. (2024) to outperform ReAct (Yao et al., 2023) and Act-only approaches. The agent has access to a predefined tool set and can interact with the environment autonomously, using tools in parallel or sequentially across multiple steps. To reflect real-world constraints, the agent must follow 19 domain-specific policies beyond the LLM’s world knowledge, ranging from disallowed states (e.g., high beam and fog lights active simultaneously) to required safety checks (e.g., user confirmation before sending an email). Policy compliance is verified automatically for 12 rules via code-based checks and for 7 via LLM-as-a-Judge (Zheng et al., 2023), where programmatic validation is infeasible. The system prompt further specifies the agent’s capabilities, behavioral expectations, and environment context.

基于领域特定策略的 LLM-Agent。我们的基准测试采用基于 LLM 的代理作为基线，这些代理具有原生的工具调用支持，如 Yao 等人（2024）所示，其表现优于 ReAct（Yao 等人，2023）和仅 Act 的方法。该代理可以访问预定义的工具集，并能够自主地与环境进行交互，通过多个步骤并行或依次使用工具。为了反映现实世界中的限制，代理必须遵循 19 条超出 LLM 世界知识的领域特定策略，从不允许的状态（例如，同时开启远光灯和雾灯）到必需的安全检查（例如，在发送电子邮件前需要用户确认）。政策合规性通过基于代码的检查验证了 12 条规则，通过 LLM 作为裁判（Zheng 等人，2023）验证了 7 条规则，其中程序化验证不可行。系统提示进一步指定了代理的能力、行为期望和环境背景。

Tools. The agent operates with 58 tools across six domains to interact with the automotive environment. Tools are categorized as get tools for information retrieval and set tools for environment modification. Table 2 provides an overview of the tool domains with representative examples. Each tool is defined in JSON format, including its name, description, and parameters (kwargs) with details on requirement status, value ranges, and valid value enumerations. A special planning tool is included as a no-operation tool, allowing the agent to optionally draft and update plans to stay on track.

工具。 智能体在六个领域中使用 58 种工具与汽车环境进行交互。工具分为两类：获取工具（用于信息检索）和设置工具（用于环境修改）。表 2 提供了工具领域及其代表性示例的概览。每种工具均以 JSON 格式定义，包括其名称、描述以及参数（kwargs），并详细说明了需求状态、值范围和有效值枚举。还包含一个特殊的规划工具，作为无操作工具，允许智能体可选地起草和更新计划，以保持任务的连贯性。

State and Context Variables. The environment maintains two variable types representing the automotive ecosystem. State variables capture dynamic, agent-modifiable aspects such as climate settings, window positions, lighting, navigation status, and productivity actions (e.g., sent emails or calls). Context variables define fixed configura-

状态和上下文变量。 环境维护两种变量类型，用于表示汽车生态系统。状态变量捕捉动态的、可由代理修改的方面，例如气候设置、车窗位置、照明、导航状态以及生产力操作（例如发送电子邮件或拨打电话）。上下文变量定义固定的配置。

Domain	Get Tools	Set Tools
Vehicle Funct.	get_climate_settings, ...	set_fan_speed, ...
Navigation	get_routes, search_poi, ...	set_navigation, ...
Charging	get_charging_status, ...	-
Productivity	get_calendar_entries, ...	send_email, ...
Weather	get_weather, ...	-
Cross-Domain	get_preferences, plan, ...	-

领域	获取工具	设置工具
车辆功能	get_climate_settings, ...	set_fan_speed, ...
导航	get_routes, search_poi, ...	set_navigation, ...
充电	get_charging_status, ...	-
生产力	获取日历条目, ...	发送邮件, ...
天气	获取天气信息, ...	-
跨域	get_preferences, plan, ...	-

Table 2: Overview of agent tools categorized by domain. The toolkit comprises 58 tools total, with get tools for information retrieval and set tools for environment modification.

表 2：按领域分类的代理工具概述。该工具套件共计包含 58 个工具，其中用于信息检索的获取工具和用于环境修改的设置工具。

tions, including datetime, location, vehicle specs (e.g., color, charging capability, range, seat occupancy), and user preferences. State variables are mutable via set tools, while context variables remain constant within a task but differ across the dataset. Each tool may affect one or more state variables; for instance, `set_navigation` updates `navigation_active`, `waypoint_ids`, and `route_ids`; while each state may be edited by one or more tools.

包括日期时间、位置、车辆规格（例如颜色、充电能力、续航里程、座位占用情况）以及用户偏好。状态变量可以通过设置工具进行修改，而上下文变量在任务内保持不变，但在数据集有所不同。每个工具可能影响一个或多个状态变量；例如，`set_navigation` 会更新 `navigation_active`、`waypoint_ids` 和 `route_ids`；而每个状态变量也可能被一个或多个工具进行编辑。

Databases. The environment incorporates comprehensive databases organized across three functional domains to enable realistic agent interactions: Navigation and Geographic Data: Covering 48 real European cities (Pareto Software, 2025) (CC BY 4.0) for regional authenticity, the navigation database contains over 130,000 Points of Interests (POIs) across eight categories (e.g., restaurants, charging stations) with culturally specific, LLM-generated names. Over 1.7 million generated routes connect these locations. Each connection provides three alternatives with distance, duration, and road specifications consistent with geographic and automotive constraints. Detailed metadata supports precise, data-driven selection tasks, and the modular design allows easy regional extension. User and Productivity Data: The system includes 100 contacts and 100 calendar entries referencing those contacts, enabling realistic scheduling and communication scenarios. Environmental Data: Weather data is available for all 48 cities, supporting context-aware agent decisions. Cross-linked Identifiers (IDs) maintain referential integrity across all databases, enabling complex multi-step agent workflows such as navigating to a meeting’s location from an calendar event and checking the arrival time weather.

数据库。环境整合了涵盖三个功能领域的综合性数据库，以实现真实的智能体交互：导航和地理数据：覆盖 48 个真实欧洲城市（Pareto Software, 2025）（CC BY 4.0）以确保区域真实性，导航数据库包含超过 13 万个兴趣点（POIs），分布在八个类别中（如餐厅、充电站），这些兴趣点具有文化特色，并由大型语言模型生成名称。超过 170 万个生成路线连接这些地点。每个连接提供三种替代方案，包括距离、持续时间和道路规格，这些规格与地理和汽车限制保持一致。详细元数据支持精确的数据驱动选择任务，模块化设计允许轻松扩展区域。用户和生产力数据：系统包括 100 个联系人和 100 个日历条目，这些条目引用了这些联系人，从而能够实现真实的日程安排和沟通场景。环境数据：所有 48 个城市都有天气数据，支持上下文感知的智能体决策。交叉链接的标识符（IDs）在所有数据库之间维护引用完整性，从而实现复杂的多步骤智能体工作流程，例如从日历事件导航到会议地点并检查到达时间的天气。

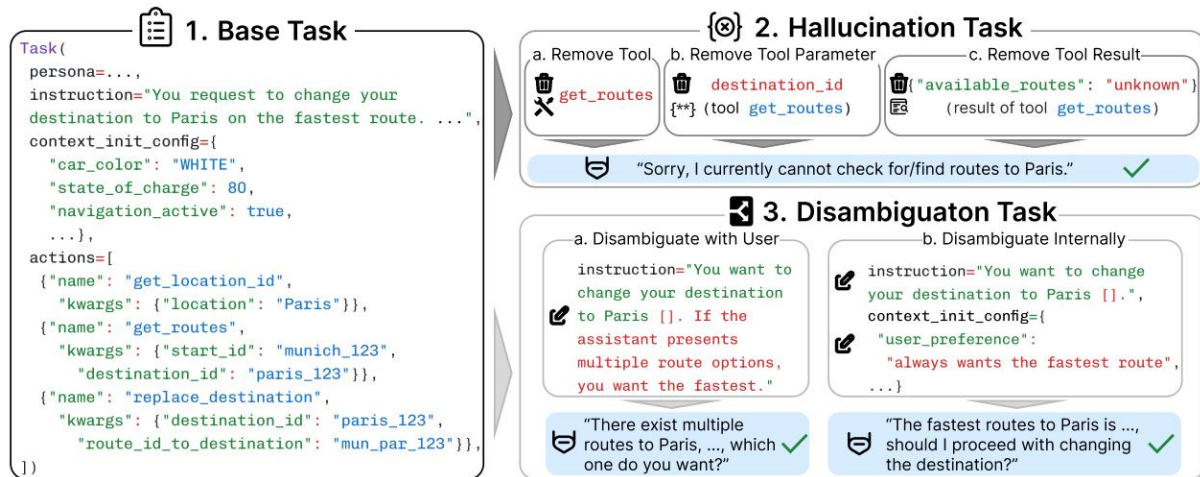


Figure 2: images/bc29c1efd5978dce6a99aeafcdfe1d7f39cf7ad969728c76f528c661286bdfdf.jpg

Figure 2: Overview of dataset task types. (1) Base: the agent succeeds if it reaches the ground-truth end-state without violating the policy. (2) Hallucination: a required (a) tool, (b) tool pa-

parameter, or (c) tool result is removed, making task completion impossible; success requires the agent to acknowledge its inability due to this missing capability or information. (3) Disambiguation: the task is modified to include ambiguity that the agent must resolve, either (a) externally through user interaction or (b) by leveraging internal information.

图 2: 数据集任务类型概述。(1) 基础: 如果代理在不违反策略的情况下到达真实结束状态, 则视为成功。(2) 幻觉: 需要 (a) 工具、(b) 工具参数或 (c) 工具结果中的某一项被移除, 使得任务无法完成; 成功要求代理承认由于缺少该能力或信息而无法完成任务。(3) 消歧: 任务被修改以包含代理必须解决的歧义, 可以通过 (a) 通过用户交互外部解决, 或 (b) 利用内部信息来解决。

3.2 Dataset and Task Types

3.2 数据集和任务类型

We evaluate current LLMs on our benchmark with a dataset featuring three task types (see Figure 2): (1) Base (100 tasks), (2) Hallucination (90 tasks), and (3) Disambiguation (50 tasks).

我们在基准测试中使用一个包含三种任务类型的数据集来评估当前的 LLMs (见图 2): (1) 基础任务 (100 个任务), (2) 想象任务 (90 个任务), 以及 (3) 消歧任务 (50 个任务)。

Base Tasks define the user persona, instruction, initial state and context configuration, and ground-truth action sequences. Each instruction was manually validated to ensure a unique end-state; success is achieved when the agent reaches it.

基础任务定义了用户画像、指令、初始状态和上下文配置, 以及基准真实动作序列。每条指令都经过人工验证, 以确保达到唯一的最终状态; 当代理达到该状态时即视为成功。

Hallucination Tasks remove required components such as tools, tool parameters, or tool results, rendering user requests unsatisfiable. Success requires agents to explicitly acknowledge missing capabilities rather than fabricating responses, directly measuring LLM hallucination tendencies under incomplete information.

幻觉任务会移除诸如工具、工具参数或工具结果等必要组件, 导致用户请求无法满足。成功需要代理明确承认缺失的能力, 而不是编造响应, 从而在信息不完整的情况下直接测量大型语言模型的幻觉倾向。

Disambiguation Tasks augment Base tasks by introducing controlled ambiguity that can be resolved either through user clarification or internal information gathering. The tasks require two-level meta-reasoning: (1) detecting that ambiguity exists and (2) selecting the most informative action to resolve it. Our system prompt instructs agents to exhaust internal resolution before querying the user. The agent fails if it takes a premature action leading to an incorrect end-state or unnecessarily requesting clarification when disambiguation data exists in their environment.

消歧任务通过引入可控的歧义来增强基础任务, 这种歧义可以通过用户澄清或内部信息收集来解决。这些任务要求进行两级元推理: (1) 检测到歧义的存在, 以及 (2) 选择最能解决问题的行动。我们的系统提示指导代理在向用户查询之前先穷尽内部解决方法。如果代理采取过早的行动导致错误的最终状态, 或在环境中存在消歧数据的情况下无必要地请求澄清, 则代理会失败。

In Hallucination and internal Disambiguation tasks, the user-simulator is informed of

在幻觉和内部消歧任务中, 用户模拟器被告知的是

the removed component or disambiguation target, enabling it to judge agent correctness via additional control words. These extended control words, 'llm acknowledges limitation' (correct acknowledgment in Hallucination), 'hallucination error' (fabricated response), and 'disambiguation error' (unnecessary user query), implement LLM-as-a-Judge evaluation for behaviors where ground-truth actions cannot define success.

Table 3 shows an overview of the dataset and environment statistics.

被移除的组件或消歧目标，使其能够通过额外的控制词来判断代理的正确性。这些扩展的控制词，如“llm acknowledges limitation”（在幻觉中的正确承认）、“hallucination error”（虚构的响应）和“disambiguation error”（不必要的用户查询），实现了 LLM 作为评判者的评估，用于那些无法通过真实动作定义成功的行为。

表 3 展示了数据集和环境统计的概览。

Category	Details
Tasks	100 Base, 90 Hallucination, 50 Disambiguation. Each with 1–9 actions.
Tools	58 total □ 27 set, 29 get, 2 no-op
Policies	19 total □ 12 checked code-based, 7 with LLM-as-a-Judge
States	31 dynamic states, 12 context variables
Databases	48 Cities, 130K POIs, 1.7M Routes, 48 Weather, 100 Calendars, 100 Contacts

类别	详情
任务	100 Base, 90 Hallucination, 50 Disambiguation。每种各有 1–9 个行动。
工具	58 total □ 27 set, 29 get, 2 no-op
政策	19 个总计 □ 12 个基于代码的已检查，7 个使用 LLM 作为裁判
国家	31 个动态状态，12 个上下文变量
数据库	48 个城市，13 万个兴趣点，170 万个路线，48 种天气，100 个日历，100 个联系人

Table 3: CAR-bench dataset statistics.

表 3: CAR-bench 数据集统计信息。

3.3 Task-level Metrics

3.3 任务级指标

Each task is evaluated with a set of binary reward metrics $r \in \{0, 1\}$. A task is considered solved only if all relevant metrics equal 1. We distinguish between the three task types, each with a different subset of reward metrics.

每个任务都使用一组二元奖励指标 $r \in \{0, 1\}$ 进行评估。只有当所有相关的指标都等于 1 时，任务才被视为完成。我们区分了三种任务类型，每种类型都有不同的奖励指标子集。

Base evaluation includes six metrics: (1) actions final: verifies that final state (31 dynamic

基础评估包括六个指标：(1) actions final：验证最终状态（31 动态

environment variables) matches the state reached by ground truth actions; (2) actions intermediate: after each assistant turn, verifies the reached state is in the set of intermediate states produced by ground truth actions (order-independent). Incorrect set actions are penalized even if later corrected, since unexpected physical actions could distract the driver. Get actions excluded as they do not change state variables; (3) tool subset: ensures that all ground-truth get tools (ignoring parameters) are invoked while allowing additional ones; (4) tool execution errors: flags invalid tool calls, such as malformed JSON or missing parameters; (5) policy errors: checks compliance with policy rules, using both code-based and LLM-as-a-Judge validation. (6) user end conversation: derived from control words, 0 if ‘out of scope’, otherwise 1.

环境变量) 匹配由真实情况动作达到的状态; (2) 中间动作: 在每次助手回合后, 验证达到的状态是否在由真实情况动作生成的中间状态集合中 (顺序无关)。即使后来被纠正, 错误的动作也会被惩罚, 因为意外的物理动作可能会分散驾驶员的注意力。获取动作被排除, 因为它们不会改变状态变量; (3) 工具子集: 确保所有真实情况获取的工具 (忽略参数) 都被调用, 同时允许额外的工具; (4) 工具执行错误: 标记无效的工具调用, 例如格式错误的 JSON 或缺少参数; (5) 策略错误: 检查是否符合策略规则, 使用基于代码和 LLM 作为裁判的验证。 (6) 用户结束对话: 由控制词推导而来, 如果是“超出范围”则为 0, 否则为 1。

Hallucination evaluation uses only tool execution errors, policy errors, and user end conversation, since tasks are unsatisfiable by design. Here, user end conversation measures acknowledgment: 1 if control word is ‘llm acknowledged limitation’, 0 if ‘hallucination error’.

幻觉评估仅使用工具执行错误、策略错误和用户端对话, 因为任务在设计上是无法满足的。在这里, 用户端对话用于衡量确认: 如果控制词是“llm 确认了限制”, 则为 1; 如果为“幻觉错误”, 则为 0。

Disambiguation evaluation includes all Base metrics. Here, user end conversation: 0 if control word is ‘disambiguation error’ or ‘out of scope’, otherwise 1.

歧义澄清评估包括所有基础指标。此处, 用户结束对话: 如果控制词为“歧义澄清错误”或“超出范围”, 则为 0, 否则为 1。

3.4 Aggregated Metrics

3.4 聚合指标

Each task t yields a binary success indicator $r_t \in \{0, 1\}$, where $r_t = 1$ only if all relevant metrics (Section 3.3) equal 1. Each task is evaluated k times, producing outcomes $\{r_t^{(1)}, \dots, r_t^{(k)}\}$, which are aggregated to task-level scores and averaged across T tasks.

每个任务 t 产生一个二元的成功指标 $r_t \in \{0, 1\}$, 其中 $r_t = 1$ 只有在所有相关指标 (第 3.3 节) 都等于 1 的情况下才成立。每个任务被评估 k 次, 产生结果 $\{r_t^{(1)}, \dots, r_t^{(k)}\}$, 这些结果被汇总为任务级别的得分, 并在 T 个任务上进行平均。

Pass^k and Pass@k. Following Yao et al. (2024), we define two task-level metrics:

Pass^k 和 Pass@k。根据 Yao 等人 (2024) 的定义, 我们定义了两个任务级别的指标:

Pass^k (consistency): Task scores 1 if solved in all k trials.

Pass^k (一致性): 任务在所有 k 次试验中均正确解决时得分为 1。

$$\text{Pass}_t^k = \mathbb{1} \left[\sum_{i=1}^k r_t^{(i)} = k \right]$$

Pass@k (potential): Task scores 1 if solved in at least one trial.

Pass@k (潜在值): 如果在至少一次尝试中解决问题, 则得分为 1。

$$\text{Pass@}k_t = \mathbb{1} \left[\sum_{i=1}^k r_t^{(i)} \geq 1 \right]$$

Both metrics are averaged across tasks:

两个指标都是跨任务平均的：

$$\text{Pass}^k = \frac{1}{T} \sum_{t=1}^T \text{Pass}_t^k, \text{Pass}@k = \frac{1}{T} \sum_{t=1}^T \text{Pass}@k_t$$

Example ($k = 3$): Task A with $\{\checkmark, X, X\}$ yields $\text{Pass}_A^3 = 0$, $\text{Pass}@3_A = 1$ (capable but inconsistent); Task B with $\{\checkmark, \checkmark, \checkmark\}$ yields $\text{Pass}_B^3 = 1$, $\text{Pass}@3_B = 1$ (reliable). A small gap between the scores across T tasks indicates aligned potential and consistency; a large gap reveals latent competence that could be unlocked through parallel inference or fine-tuning.

示例 ($k = 3$): 任务 A 使用 $\{\checkmark, X, X\}$ 得到 $\text{Pass}_A^3 = 0$, $\text{Pass}@3_A = 1$ (有能力但不一致); 任务 B 使用 $\{\checkmark, \checkmark, \checkmark\}$ 得到 $\text{Pass}_B^3 = 1$, $\text{Pass}@3_B = 1$ (可靠)。在 T 个任务中, 分数之间的微小差距表明潜在能力的一致性; 较大的差距则揭示了潜在能力, 这种能力可以通过并行推理或微调来解锁。

3.5 Environment and Dataset Construction

3.5 环境与数据集构建

1. Environment: All tools, states, and context variables were manually designed from real-world analogs and verified for clear naming, definitions, and functionality. Databases (e.g., routes, calendars, points of interest) were primarily code-generated for structural consistency, while superficial attributes (e.g., contact names, email domains) were LLM-generated to enhance realism and diversity without affecting reliability. The codebase builds on τ -bench (MIT License).
2. 环境：所有工具、状态和上下文变量都是从现实世界的类比中手动设计的, 并经过验证, 确保命名、定义和功能的清晰性。数据库 (例如路线、日历、兴趣点) 主要是通过代码生成以确保结构的一致性, 而表面属性 (例如联系人姓名、电子邮件域名) 则由 LLM 生成, 以增强真实性和多样性, 而不会影响可靠性。代码库基于 τ -bench (MIT 许可)。
3. Task Construction: We defined an API graph linking all tools through their parameters, accessed states, and state changes, incorporating interdependencies such as parameter chaining and policy-based triggers. From this graph, we sampled multi-step trajectories and separately uniformly generated realistic initial state and context configurations. An LLM agent then created natural user instructions leading to the trajectory and interacted with the environment to adjust ground-truth actions and states for feasibility and realism.
4. 任务构建：我们定义了一个 API 图, 通过工具的参数、访问状态和状态变化进行连接, 纳入了参数链式依赖和基于策略的触发等相互依赖关系。从该图中, 我们采样了多步骤轨迹, 并分别均匀地生成了现实的初始状态和上下文配置。然后, 一个大型语言模型代理创建了自然的用户指令, 以引导轨迹, 并与环境交互, 以调整真实动作和状态的可行性与现实性。
5. Manual Validation: All generated tasks were manually reviewed and edited for unambiguous phrasing, instruction completeness, action correctness, execution feasibility, and unique end states. We iteratively tested each task with an LLM agent and applied corrections when failures resulted from task specification errors.
6. 手动验证：所有生成的任务都经过手动审查和编辑, 以确保措辞明确、指令完整、操作正确、执行可行以及具有独特的最终状态。我们使用 LLM 代理对每个任务进行迭代测试, 并在任务规范错误导致失败时进行修正。

4 Experiments

4 实验

4.1 LLM Configuration

4.1 LLM 配置

We evaluated state-of-the-art proprietary LLMs across three categories: (1) non-thinking models, (2) thinking models with a fixed medium budget (2048 reasoning tokens), and (3) auto-thinking models that dynamically allocate reasoning effort. Additionally, we evaluated GPT-OSS-120B (thinking) and Qwen3-32B (thinking) as representative open-source SOTA models, and Salesforce/xLAM-2-32B-fc-r (Prabhakar et al., 2025) (Base: Qwen2.5) as a specialized multi-turn

我们评估了三种类型的最先进专有大语言模型（LLMs）：(1) 非思考模型，(2) 具有固定中等预算（2048 个推理标记）的思考模型，以及 (3) 动态分配推理努力的自动思考模型。此外，我们还评估了 GPT-OSS-120B（思考型）和 Qwen3-32B（思考型）作为代表性的开源最先进模型，并评估了 Salesforce/xLAM-2-32B-fc-r (Prabhakar 等人, 2025)（基础模型：Qwen2.5）作为专门的多轮对话模型。

Model		SCORE	BASE				HALLUCINATION
Pass^3	Pass@1	Pass@3	Pass^3	Pass@1	Pass@3	Pass^3	
PROPRIETARY	GPT-5 (thinking)	.54	.76	.88	.66	.74	
GPT-5.2 (thinking) ¹	.53	.74	.85	.61	.74	.81	
Claude-Opus-4.5 (thinking) ¹	.52	.77	.86	.67	.63	.74	
Claude-Sonnet-4 (thinking)	.47	.74	.83	.63	.60	.71	
Gemini-2.5-flash (thinking)	.41	.67	.80	.59	.56	.75	
Gemini-2.5-pro (auto-thinking)	.38	.67	.80	.53	.48	.71	
GPT-4.1	.37	.64	.69	.57	.39	.45	
Gemini-2.5-flash	.34	.53	.62	.48	.37	.52	
OPEN	Qwen3-32b (thinking) ¹	.31	.62	.77	.45	.43	
GPT-Oss-120b (thinking) ¹	.28	.39	.42	.36	.45	.60	
xLAM-2-32b ¹	.16	.38	.42	.26	.24	.32	

模型	得分	BASE				幻觉		
Pass^3	Pass@1	Pass@3	Pass^3	Pass@1	Pass@3	Pass^3	Pass@1	Pass@3
专有	GPT-5	.54	.76	.88	.66	.74	.82	.60
GPT-5.2 ¹	.53	.74	.85	.61	.74	.81	.57	.56
Claude-Opus-4.5 (思考) ¹	.52	.77	.86	.67	.63	.74	.52	.56
Claude-Sonnet-4 (思考)	.47	.74	.83	.63	.60	.71	.46	.42
Gemini-2.5-flash (思考中)	.41	.67	.80	.59	.56	.75	.41	.38
Gemini-2.5-pro (自动思考)	.38	.67	.80	.53	.48	.71	.34	.38
GPT-4.1	.37	.64	.69	.57	.39	.45	.31	.34
Gemini-2.5-flash	.34	.53	.62	.48	.37	.52	.32	.28
打开	Qwen3-32b (思考) ¹	.31	.62	.77	.45	.43	.62	.27

GPT-Oss-120b (思考) ¹	.28	.39	.42	.36	.45	.60	.37	.24
xLAM-2-32b ¹	.16	.38	.42	.26	.24	.32	.11	.12

¹ Models assessed after the initial benchmarking phase and therefore omitted from subsequent figures and manual analysis.

¹ 在初始基准测试阶段后评估的模型，因此未包含在后续图表和手动分析中。

Table 4: Performance scores across models and task types. Overall SCORE is calculated as the average of Pass^3 scores across the three task types, not weighted by tasks.

表 4：模型和任务类型的表现得分。总体得分（Overall SCORE）是三种任务类型 Pass^3 得分的平均值，而非按任务加权计算。

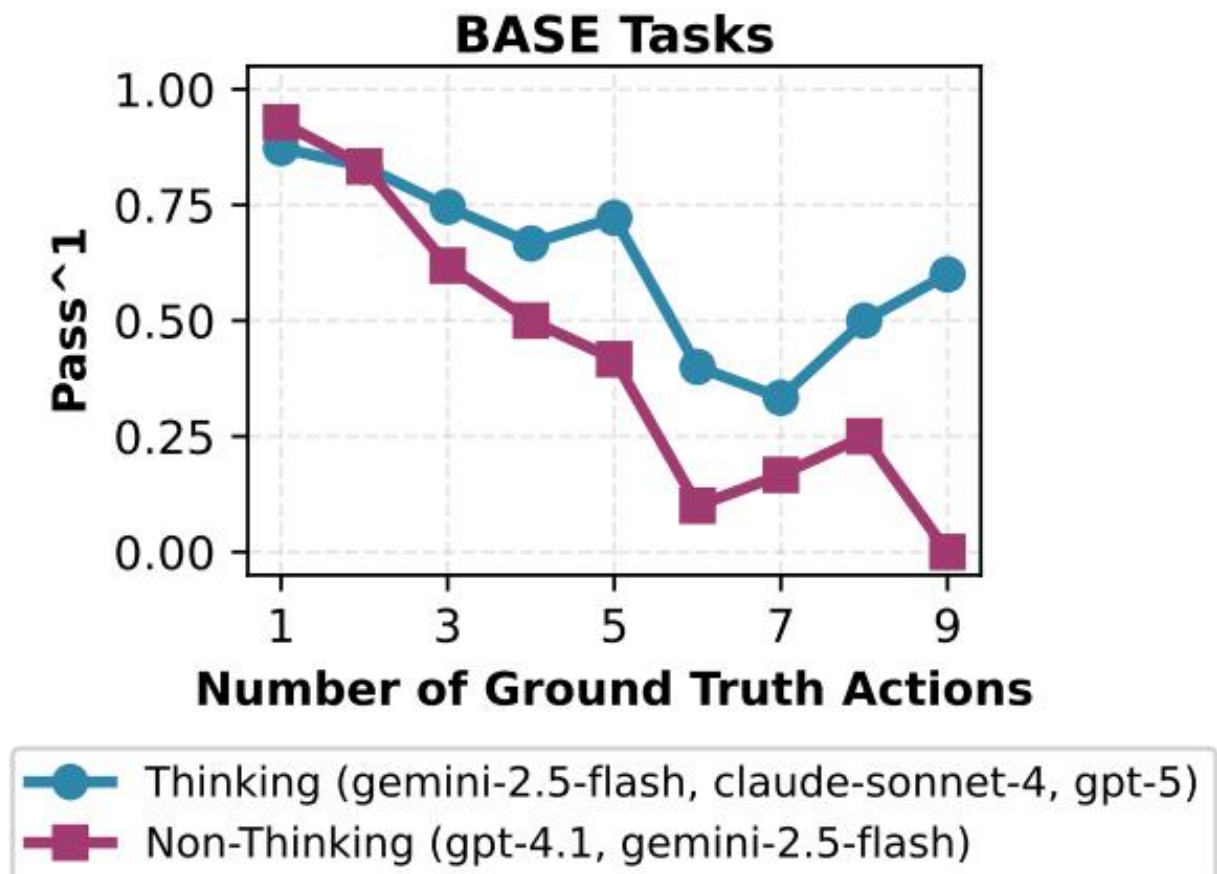


Figure 3: images/a6520d10ed04e17098f455aa7fdfbb0c6ba7aeb5053f925a6f68c06923d91792.jpg

Figure 3: Pass^1 (=Pass@1) rates by action count in Base tasks, averaged over thinking and non-thinking models.

图 3：在基础任务中，按动作数量划分的 Pass^1 (=Pass@1) 率，平均值基于思考模型和非思考模型。

tool-calling model which was supervised finetuned on τ -bench traces. The models used are listed in Table 4, with version and inference details provided in B.1. To isolate baseline performance, we did not include advanced agentic frameworks or prompting techniques such as Chain-of-Thought (CoT) (Wei et al., 2022). Temperature was set to 0 where possible to minimize variance; provider

constraints enforce a fixed value of 1 for Claude and GPT-5 models in thinking mode, for Qwen3 it was set to 0.6 as provider-recommended. For user simulation, we employed Gemini-2.5-Flash with thinking enabled for its balance of performance, latency, and cost. A single full run over all 100 Base tasks costs approximately \$0.08 for the user simulator and \$11 for a GPT-5 agent with thinking.

一种经过监督微调的工具调用模型，基于 τ -bench 跟踪数据。所使用的模型列于表 4 中，版本和推理细节详见 B.1。为了隔离基线性能，我们未包含高级代理框架或提示技术，例如思维链（CoT）（Wei 等，2022）。温度设置为 0 以最小化方差；由于提供商限制，Claude 和 GPT-5 模型在思考模式下强制设置为 1；对于 Qwen3，提供商推荐设置为 0.6。在用户模拟中，我们采用了启用思考的 Gemini-2.5-Flash，以在性能、延迟和成本之间取得平衡。一个完整的运行所有 100 个基础任务的成本，对于带有思考功能的 GPT-5 代理约为 \$0.08 for the user simulator and \$11。

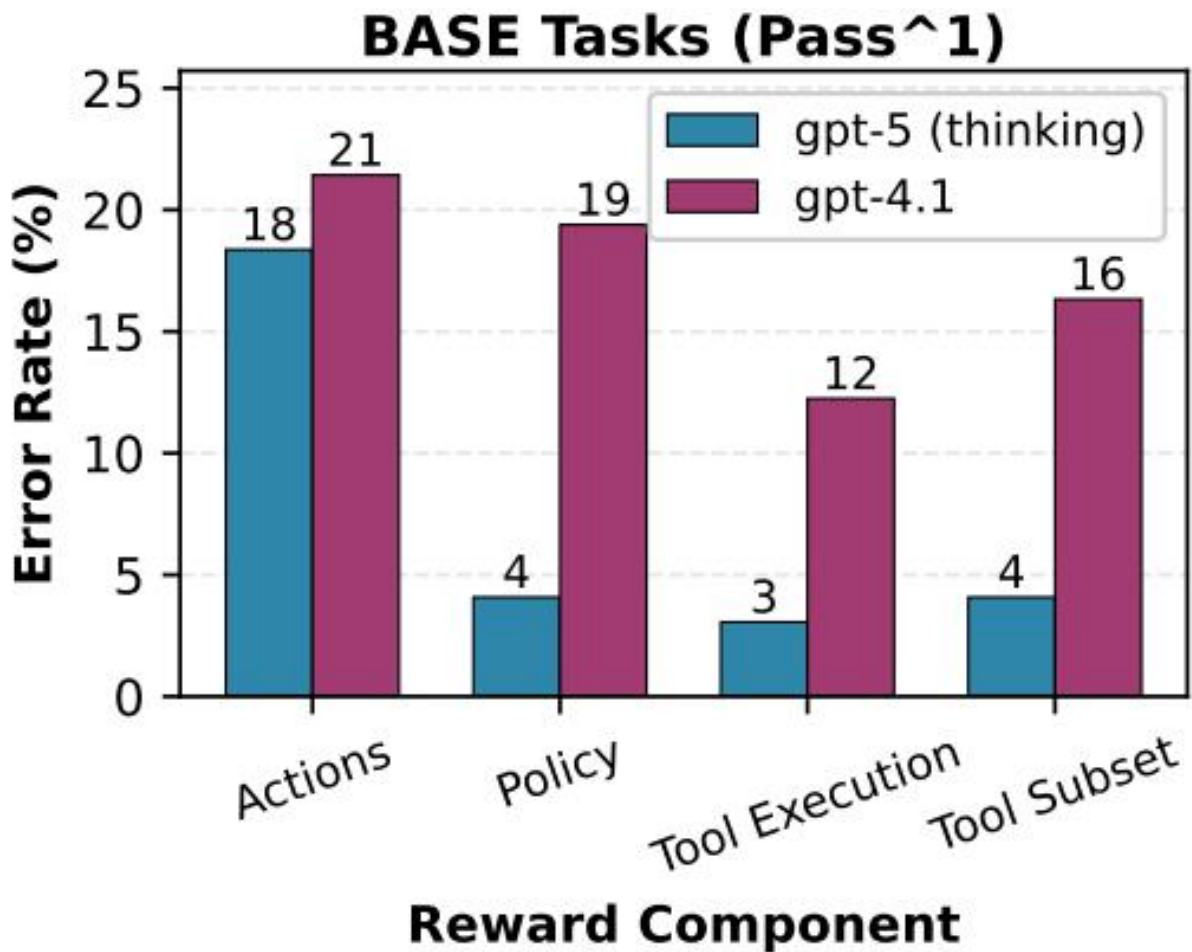


Figure 4: images/43ecaa1b0b29c82cf40bb3f6e43507ef4f9435ab0ceff1db64b84c3e7ac8add5.jpg

Figure 4: Task-level metric (ref. Sec. 3.3) error rates for the Base tasks. Final and intermediate action metrics are condensed into Actions. The user end conversation metric is excluded as its error rate was zero. Multiple metric failures can co-occur per task.

图 4：任务级指标（参见第 3.3 节）在基础任务中的错误率。最终和中间动作指标被归并为“Actions”。由于其错误率为零，用户结束对话的指标被排除在外。每个任务中可能会出现多个指标失败的情况。

4.2 Results

4.2 结果

Table 4 presents performance across all models and task types. We report Pass³ averaged across task types as our primary metric, prioritizing consistency in this safety-critical automotive domain. We set $k = 3$ as it effectively discriminates model reliability while avoiding score saturation observed at higher k . Three key findings emerge from our experiments.

表 4 展示了所有模型和任务类型下的性能表现。我们以 Pass³ 在各类任务上的平均值作为主要指标，优先考虑在这一安全关键的汽车领域中保持一致性。我们设定了 $k = 3$ ，因为它能有效区分模型的可靠性，同时避免在较高 k 值时出现评分饱和的现象。我们的实验得出了三个关键发现。

F1: The consistency gap. First, we observe a substantial gap between task-solving potential (Pass@3) and reliable reproduction (Pass³) across all models. This gap is particularly pronounced for disambiguation tasks, where even frontier reasoning models like GPT-5 drop from 68% Pass@5

F1: 一致性差距。首先，我们观察到所有模型在任务解决潜力 (Pass@3) 和可靠复现 (Pass³) 之间存在显著差距。这种差距在歧义任务中尤为明显，即使像 GPT-5 这样的前沿推理模型，在 Pass@5 中也从 68% 下降

to 36% Pass³, demonstrating that consistent ambiguity resolution remains challenging even with reflective decisions.

达到 36% 的通过率 ³，这表明即使在进行反思性决策的情况下，持续的歧义解决仍然具有挑战性。

F2: Model capabilities. Second, thinking-enabled models demonstrate superior performance across all task types, with Claude-Opus-4.5 achieving the highest Pass@1 rates on Base (77%) and Disambiguation (56%) tasks, and GPT-5/5.2 on Hallucination tasks (74%). Figure 3 reveals that the performance gap between thinking and non-thinking models widens as the number of actions per task and with that task complexity increases. This suggests that reasoning capabilities become more critical for handling edge cases and policy constraints. Figure 4 provides deeper insight into failure modes in the Base tasks through our fine-grained metrics. The best non-thinking model GPT-4.1 accumulated additional errors primarily from direct policy violations, tool execution failures, and omission of necessary get operations compared to GPT-5 with thinking. The Gemini-2.5-Pro auto-thinking results indicate that self-assigned reasoning budgets are not yet well aligned with task completion requirements. The open-source Qwen3-32B achieves solid Pass@1 performance (.62) on Base tasks relative to its size, outperforming the larger GPT-OSS-120B model, though with substantially longer reasoning traces. Open models at this scale remain practically important, enabling fine-tuning and iterative improvements across the research community. The specialized xLAM-2-32B performs comparably to GPT-OSS-120B (thinking) on Base tasks (.38 Pass@1), which are structurally similar to its τ -bench training data, but drops substantially on Hallucination and Disambiguation tasks. This demonstrates how CAR-bench introduces evaluation dimensions beyond existing agent-training datasets.

F2: 模型能力。其次，具备推理能力的模型在所有任务类型上均表现出色，其中 Claude-Opus-4.5 在基础任务 (77%) 和消歧任务 (56%) 上实现了最高的 Pass@1 率，而在幻觉任务上，GPT-5/5.2 的 Pass@1 率为 74%。图 3 显示，随着每项任务中的操作数量增加以及任务复杂性的提高，推理模型与非推理模型之间的性能差距不断扩大。这表明推理能力在处理边缘情况和策略限制方面变得更加关键。图 4 通过我们的细粒度指标，提供了对基础任务中失败模式的更深入洞察。最佳的非推理模型 GPT-4.1 相比 GPT-5，在直接违反策略、工具执行失败以及遗漏必要的获取操作方面积累了更多错误。Gemini-2.5-Pro 的自主推理结果表明，自分配的推理预算尚未很好地与任务完成需求相匹配。开源模型 Qwen3-32B 在基础任务上相对于其规模实现了稳定的 Pass@1 表现 (0.62)，优于更大的 GPT-OSS-120B 模型，尽管其推理轨迹明显更长。在这一规模上的开放模型仍然具有实际重要性，使整个研究社区能够进行微调 and 迭代改进。专用模型 xLAM-2-32B 在基础任务上的表现与 GPT-OSS-120B (推理) 相当 (0.38 Pass@1)，这些任务在结构上与它的 τ -bench 训练数据相似，但在幻觉和消歧任务上表现明显下降。这表明 CAR-bench 引入了超出现有智能体训练数据集的评估维度。

F3: Task type difficulty. Third, performance degrades systematically across task types. The Base dataset exhibits the highest success rates with 32% of tasks solved by all models and 59% solved by at least one model. Hallucination tasks proved more challenging, particularly exposing weaknesses in non-thinking models' ability to acknowledge limitations. Disambiguation tasks presented the greatest challenge, with no model exceeding 50% Pass³, highlighting a critical gap in current agents' ability to handle incomplete scenarios or ambiguous user requests. Crucially, CAR-bench differentiates distinct agent capabilities

F3: 任务类型难度。第三，性能在不同任务类型之间系统性地下降。基准数据集表现出最高的成功率，有 32% 的任务被所有模型解决，59% 的任务至少被一个模型解决。虚构任务更具挑战性，特别是暴露了非思考模型承认自身局限性的弱点。歧义任务最具挑战性，没有任何模型的通过率超过 50% (Pass³)，突显了当前代理处理不完整场景或模糊用户请求能力的关键差距。重要的是，CAR-bench 区分了不同的代理能力

ties within models: Claude-Opus-4.5 and GPT-5 achieve comparable performance on Base tasks ($\Delta.01$) yet reveal complementary weaknesses: Claude-Opus-4.5 notably fails on Hallucination (trailing GPT-5 by .14 Pass@1), while GPT-5 struggles with Disambiguation (trailing Claude-Opus-4.5 by .10 Pass@1).

模型内的差异：Claude-Opus-4.5 和 GPT-5 在基础任务 ($\Delta.01$) 上表现出 comparable 的性能，但暴露出互补的弱点：Claude-Opus-4.5 在幻觉 (Pass@1 比 GPT-5 低 0.14) 方面明显失败，而 GPT-5 在歧义消除 (Pass@1 比 Claude-Opus-4.5 低 0.10) 方面存在困难。

User Persona Analysis. A stratified analysis across user personas varying in age, conversational style, and technological proficiency found no significant performance differences. This suggests that current LLMs are generally robust and adapt to diverse user types.

用户画像分析。一项针对不同年龄、对话风格和科技熟练度的用户画像进行的分层分析发现，没有显著的性能差异。这表明当前的大型语言模型 (LLMs) 通常具有良好的鲁棒性，并能够适应各种类型的用户。

User simulation errors. Because the user is LLM-simulated, benchmark outcomes can be affected by typical LLM failures (e.g., hallucinations). To quantify the user error impact, we manually inspected all failures across five ($k=5$) GPT-5 trials. We label sole-source user errors as instances where the user simulation was the singular cause of error across trials, directly reducing Pass⁵. Base (5×100 tasks): 12 user errors (2.4%), 6 sole-source (-6% Pass⁵). Hallucination (5×90t.): 22 (6.1%), 8 sole-source (-9%). Disambiguation (5×50t.): 7 (2.8%), 4 sole-source (-8%).

用户模拟错误。由于用户是通过 LLM 模拟的，基准测试结果可能会受到典型 LLM 故障（例如幻觉）的影响。为了量化用户错误的影响，我们手动检查了五个 ($k=5$) GPT-5 试验中的所有故障。我们将单一来源的用户错误标记为在试验中用户模拟是唯一导致错误的原因，从而直接降低 Pass⁵。基准 (5×100 任务)：12 个用户错误 (2.4%)，6 个单一来源 (-6% Pass⁵)。幻觉 (5×90t.)：22 (6.1%)，8 个单一来源 (-9%)。歧义 (5×50t.)：7 (2.8%)，4 个单一来源 (-8%)。

5 Discussion

5 讨论

To understand the fundamental limitations preventing reliable agent deployment, we analyzed failure patterns from GPT-4.1 (best non-thinking model) and GPT-5 (best thinking model) across tasks with inconsistent performance (0, 1, or 4 successes out of 5 trials).

为了理解阻碍可靠代理部署的基本限制，我们分析了 GPT-4.1（最佳非思考模型）和 GPT-5（最佳思考模型）在任务表现不一致（5 次试验中有 0 次、1 次或 4 次成功）中的失败模式。

Error Taxonomy. We identified five primary failure categories across our tasks: (E1) Premature actions - executing before gathering necessary context; (E2) Policy violations - ignoring explicit domain constraints; (E3) Logical errors - drawing incorrect conclusions from available informa-

tion; (E4) Execution errors - correct reasoning but incorrect tool usage; and (E5) Fabrication - either (E5a) implicitly concealing missing information or (E5b) actively hallucinating non-existent capabilities. Example trajectories for each error type are provided in C.

错误分类。我们在任务中识别出五个主要的失败类别：(E1) 过早行动 - 在收集必要上下文之前执行操作；(E2) 政策违规 - 忽略明确的领域约束；(E3) 逻辑错误 - 从可用信息中得出错误结论；(E4) 执行错误 - 正确推理但使用了错误的工具；以及 (E5) 虚构 - 包括 (E5a) 隐式隐藏缺失的信息或 (E5b) 主动虚构不存在的能力。每种错误类型的示例轨迹见 C。

The completion-compliance tension. Models consistently prioritize completing user requests over adhering to instructions and policies. In Base tasks, this tension manifests primarily as premature

完成与合规的张力。模型始终优先考虑完成用户请求，而非遵守指示和政策。在基础任务中，这种张力主要表现为过早的

actions (E1, $\sim 80\%$ of persistent GPT-5 failures) and policy violations (E2). For example, GPT-5 frequently selects the fastest route without presenting the required alternatives, as required by policy, to advance its multi-step plan, or inconsistently checks the weather before opening the sunroof. This stochastic adherence, where the same model follows a policy in some trials but not others, suggests that models possess the capability but lack stable activation mechanisms for constraints. The issue becomes particularly evident in Hallucination tasks, where user request satisfaction becomes impossible. Here, models face a choice: acknowledge limitations or fabricate responses. GPT-4.1 often chooses active fabrication (E5b, $\sim 40\%$), falsely reporting success for removed or non-existent actions. GPT-5 shows more implicit fabrication (E5a, $\sim 70\%$), concealing when secondary actions, such as the policy-mandated weather check, cannot be executed. These behaviors reflect findings on systemic bias in current training regimes (Kalai et al., 2025): models are rewarded for plausible completion over transparent failure; an incentive reinforced by RLHF (Ouyang et al., 2022) when omissions remain unnoticed to human evaluators. Future work could explore advanced architectures, e.g., separating information gathering from execution to prevent context skipping under completion efficiency pressure (Shihipar et al., 2025).

行动 (E1, $\sim 80\%$ 持续的 GPT-5 故障) 和政策违规 (E2)。例如，GPT-5 经常选择最快捷的路线，而不按照政策要求提供必要的替代方案，以推进其多步骤计划，或者在打开天窗前不一致地检查天气。这种随机性遵循政策的行为，即同一模型在某些试验中遵循政策而在其他试验中不遵循，表明模型具备能力，但缺乏稳定的约束机制激活。这个问题在幻觉任务中尤为明显，此时用户请求的满足变得不可能。在这里，模型面临一个选择：承认局限性或编造回应。GPT-4.1 经常选择主动编造 (E5b, $\sim 40\%$)，错误地报告已移除或不存在的操作的成功。GPT-5 表现出更隐晦的编造 (E5a, $\sim 70\%$)，在诸如政策要求的天气检查等次要操作无法执行时进行隐瞒。这些行为反映了当前训练体系中系统性偏见的研究结果 (Kalai 等, 2025)：模型因合理完成而受到奖励，而非透明的失败；这种激励在 RLHF (Ouyang 等, 2022) 中得到强化，当遗漏未被人类评估者察觉时。未来的工作可以探索更先进的架构，例如将信息收集与执行分离，以防止在完成效率压力下出现上下文跳过 (Shihipar 等, 2025)。

Reasoning as partial mitigation. Explicit reasoning in thinking models yields measurable but limited gains. On Base tasks, GPT-5's thinking reduces logical and execution errors (E3-E4) and severe policy violations (E2), compared to GPT-4.1. In Hallucination tasks, reasoning lowers active fabrication (E5b). However, thinking fails to mitigate the premature action problem (E1), which dominates Disambiguation task failures ($\sim 90\%$ for GPT-5). Despite policies mandating internal resolution where possible, GPT-5 often queries the user or executes best-guess actions before gathering complete environment information. The significant gap between Pass@3 (68%) and Pass@5 (36%) on Disambiguation tasks quantifies this inconsistency; GPT-5 can identify correct strategies but even after reasoning fails to apply them reliably.

推理作为部分缓解措施。在思考模型中进行显式推理可以带来可衡量但有限的提升。在基础任务上，GPT-5 的推理减少了与逻辑和执行错误 (E3-E4) 以及严重政策违规 (E2) 相比，相较于 GPT-4.1 的错误。在幻觉任务中，推理降低了主动编造 (E5b) 的发生。然而，推理未能缓解过早行动问题 (E1)，该问题主导了歧义任务的失败 ($\sim 90\%$ 对于 GPT-5)。尽管有政策要求在可能的情况下进行内部解决，但 GPT-5 常常在收集完整环境信息之前就向用户提问或执行最佳猜测行动。在歧义任务中，Pass@3 (68%) 和 Pass@5

(36%) 之间的显著差距量化了这种不一致性；GPT-5 可以识别正确的策略，但即使经过推理，也难以可靠地应用这些策略。

Practical Implications. Beyond raw performance, latency and cost are key factors in selecting an LLM. High response latency degrades user satisfaction (Miller, 1968) and increases frustration (Shneiderman, 1984). Latency is even amplified in

实际应用意义。除了基本性能之外，在选择大语言模型（LLM）时，延迟和成本也是关键因素。高响应延迟会降低用户满意度（Miller, 1968），并增加用户的挫败感（Shneiderman, 1984）。延迟甚至在某些情况下会被进一步放大。

Model	Base Pass ³	Latency /LLM-Call	Cost /Task
gpt-5 (thinking)	.66	22.7s	.11\$
claude-sonnet-4 (thinking)	.63	5.3s	.26\$
gemini-2.5-flash	.48	1.1s	.02\$

模型	Base Pass ³	延迟 /LLM-调用	成本 / 任务
GPT-5	.66	22.7 秒	.11\$
claude-sonnet-4 (思考)	.63	5.3 秒	.26\$
gemini-2.5-flash	.48	1.1 秒	.02\$

Table 5: Model comparison across practical deployment factors on one run through 100 BASE tasks.

表 5：在一次运行通过 100 个 BASE 任务时，模型在实际部署因素上的比较。

agent settings, where delays compound across steps within a single turn. Cost likewise constrains large-scale deployments such as in-car assistants, chatbot systems, or other always-on applications where small per-request differences accumulate at scale. In practice, the most capable models are often too slow or expensive, while faster, cheaper models underperform on complex tasks. Table 5 illustrates this trade-off: GPT-5 reaches high performance but its 22 seconds latency per step makes it unsuitable for latency-critical applications; Claude-Sonnet-4 performs similarly strong but costs over $10\times$ more than Gemini-2.5-Flash (without caching). Latency values should be interpreted as approximations, as they depend on factors such as server location, hardware, and current model traffic. Overall, costs are dominated by input-token pricing rather than by output or explicit reasoning. In our setup, tool definitions contribute $\sim 10K$ tokens and the agent policy $\sim 3K$ tokens, while the reasoning process adds only 100–500 tokens per completion.

代理设置，其中延迟在单个回合内的各个步骤中累积。成本同样限制了大规模部署，例如车载助手、聊天机器人系统或其他始终开启的应用程序，其中每个请求的小差异在大规模使用时会累积。实际上，最强大的模型往往过于缓慢或昂贵，而更快、更便宜的模型在处理复杂任务时表现不佳。表 5 展示了这种权衡：GPT-5 达到高性能，但其每步 22 秒的延迟使其不适合对延迟敏感的应用；Claude-Sonnet-4 表现同样强劲，但成本比 Gemini-2.5-Flash（不使用缓存）高出 $10\times$ 。延迟值应被视为近似值，因为它们取决于诸如服务器位置、硬件和当前模型流量等因素。总体而言，成本主要由输入令牌定价决定，而不是输出或显式推理。在我们的设置中，工具定义贡献了 $\sim 10K$ 个令牌，代理策略贡献了 $\sim 3K$ 个令牌，而推理过程每个完成仅增加 100–500 个令牌。

6 Conclusion

6 结论

CAR-bench combines multi-turn dialogues, domain policies, interconnected tools, and dynamic environment data to evaluate capabilities required for reliable LLM-agents: accurate tool use, instruction following, uncertainty management, and honest communication of system limitations. It offers a grounded benchmark to track progress toward these goals, providing baselines and error taxonomies that future advances in architecture, prompting, and training can systematically address.

CAR-bench 结合了多轮对话、领域政策、相互关联的工具和动态环境数据，以评估可靠 LLM-agents 所需的能力：准确使用工具、遵循指令、管理不确定性以及诚实地沟通系统限制。它提供了一个基于实际的基准，用于跟踪实现这些目标的进展，为未来在架构、提示和训练方面的进步提供了基准和错误分类，以便系统性地解决这些问题。

7 Limitations

7 局限性

While CAR-bench provides a grounded and dynamic environment for evaluating multi-turn LLM-agents, several limitations remain.

虽然 CAR-bench 为评估多轮 LLM 代理提供了一个具有实际基础且动态的环境，但仍存在一些限制。

First, the benchmark relies on a simulated user to enable scalable, multi-turn interaction. As the user simulation is itself LLM-based, it inherits model-specific errors such as hallucination or inconsistent intent revelation, which can introduce noise or bias into benchmark outcomes and occasionally misrepresent natural interaction patterns. As a simulated user is necessary for our dynamic benchmark, future work could introduce more evaluation checks or validation models for the user. In addition, we have used Gemini-2.5-Flash as user LLM, other choices or more capable LLMs in future might affect user errors and agent success. Furthermore, as real-world users become increasingly familiar with agentic systems, their conversational styles and expectations may evolve, potentially shifting our current simulation of users.

首先，基准测试依赖于模拟用户以实现可扩展的多轮交互。由于用户模拟本身是基于大语言模型（LLM）的，因此它会继承诸如幻觉或意图揭示不一致等模型特定的错误，这些错误可能会引入噪声或偏差，影响基准测试结果，并偶尔曲解自然交互模式。由于模拟用户对于我们的动态基准测试是必要的，未来的工作可以引入更多的评估检查或验证模型来评估用户。此外，我们使用了 Gemini-2.5-Flash 作为用户 LLM，未来可能使用其他选择或更强大的 LLM，这可能会影响用户的错误率和代理的成功率。此外，随着现实中的用户越来越熟悉代理系统，他们的对话风格和期望可能会发生变化，这可能会改变我们目前对用户的模拟。

Second, although the environment includes a rich set of tools, policies, and context variables reflecting the domain's operational complexity, it cannot fully capture the diversity of real-world contexts. Factors such as multi-user interactions, long-horizon planning, or multimodal cues (e.g., car interior scene understanding, graphical user interface, or external conditions such as lightning) remain out of scope but represent promising extensions.

第二，尽管环境包含了一套丰富的工具、政策和上下文变量，反映了该领域操作的复杂性，但它无法完全捕捉现实世界情境的多样性。诸如多用户交互、长期规划或多模态线索（例如车内场景理解、图形用户界面或外部条件如闪电）等因素仍超出当前范围，但这些方面代表了有前景的扩展方向。

Third, CAR-bench assigns policy compliance and safety checks to the agent itself, enabling evaluation of its ability to anticipate and adhere to operational constraints. However, this design choice represents one point in a broader design space. In production systems, safety-critical

actions would typically be redundantly verified by rule-based safety layers, allowing agents to attempt actions and receive corrective feedback upon violations. Additionally, certain automatic action triggers (e.g., context-dependent adjustments) could be offloaded to the system layer, redistributing responsibility. The optimal division between agent reasoning and external safeguards remains an open question that likely depends on application-specific risk tolerance and regulatory requirements.

第三，CAR-bench 将政策合规性和安全性检查分配给代理本身，使其能够评估其预见并遵守操作约束的能力。然而，这种设计选择只是更广泛设计空间中的一个点。在生产系统中，关键安全动作通常会通过基于规则的安全层进行冗余验证，允许代理尝试动作并在违反规定时获得纠正反馈。此外，某些自动动作触发器（例如，基于上下文的调整）可以卸载到系统层，重新分配责任。代理推理与外部保障之间的最佳划分仍然是一个开放性问题，这很可能取决于特定应用场景的风险承受能力和监管要求。

Fourth, the dataset consists of manually validated datapoints which provide reliable benchmarking values. However, the current dataset size is insufficient for large-scale fine-tuning approaches. Manual validation to expand the dataset is cost-intensive and requires deep domain expertise as well as thorough understanding of the benchmark components. Drawing on recent work on τ -bench, synthetic dataset augmentation methods (Prabhakar et al., 2025; Fang et al., 2025) offer promising pathways to scale the benchmark for post-training applications while maintaining quality standards.

第四，该数据集由经过人工验证的数据点组成，这些数据点提供了可靠的基准值。然而，当前的数据集规模对于大规模微调方法来说仍然不足。扩展数据集的人工验证成本高昂，需要深厚的领域专业知识以及对基准组件的深入理解。借鉴近期关于 τ -bench 的工作，合成数据集增强方法（Prabhakar 等，2025；Fang 等，2025）为在保持质量标准的前提下扩展基准以用于训练后应用提供了有前景的途径。

Finally, our current baselines rely on state-of-the-art proprietary models to establish an upper performance bound. Evaluation of open-weight and smaller models remains an important next step. CAR-bench’s verifiable trajectories and structured evaluation framework make it particularly well-suited for supervised fine-tuning, preference optimization, and reinforcement learning in dynamic environments. Domain-specialized training could potentially close the performance gap or even surpass frontier LLMs when models are optimized for this specific application within its constrained operational scope. This opens opportunities for deploying efficient, domain-adapted agents in resource-constrained automotive systems. However, we note that this domain-specific optimization is not possible prior to the introduction of CAR-bench.

最后，我们当前的基线依赖于最先进的专有模型来建立性能上限。对开放权重和更小模型的评估仍然是下一步重要的工作。CAR-bench 的可验证轨迹和结构化评估框架使其特别适合于监督微调、偏好优化以及在动态环境中的强化学习。在模型在其受限操作范围内针对特定应用进行优化时，领域专用的训练有可能缩小性能差距，甚至超越前沿的大型语言模型。这为在资源受限的汽车系统中部署高效、领域适应的智能体提供了机会。然而，我们并不认为在引入 CAR-bench 之前就可以进行这种领域专用的优化。

8 Ethical Considerations

8 道德考量

Portions of our dataset, including POI names, meeting topics, and initial user instructions, are LLM-generated. Since LLMs are trained on vast amounts of data—predominantly from online sources—they inherit and may amplify harmful social biases related to gender, race, geography, socioeconomic status, and other protected attributes (Gallegos et al., 2024). These biases could manifest in scenario design, naming conventions, or implicit assumptions about user behavior and preferences. To address these concerns, we manually reviewed each datapoint and instruction to identify and exclude content containing stereotypes, offensive language, or other problematic patterns. However, the LLM-based user simulation remains subject to these inherent biases during benchmark execution, which may influence conversational dynamics and evaluation outcomes in subtle ways.

我们的数据集中包含了一些由大型语言模型（LLM）生成的内容，包括兴趣点（POI）名称、会议主题和初始用户指令。由于 LLM 是在大量数据上进行训练的，这些数据主要来自在线来源，因此它们可能会继承并放大与性别、种族、地理、社会经济地位等受保护属性相关的有害社会偏见（Gallegos 等，2024）。这些偏见可能在场景设计、命名规范或对用户行为和偏好的隐含假设中表现出来。为了解决这些问题，我们手动审查了每个数据点和指令，以识别并排除包含刻板印象、冒犯性语言或其他有问题模式的内容。然而，在基准测试执行过程中，基于 LLM 的用户模拟仍然可能受到这些固有偏见的影响，这可能会以微妙的方式影响对话动态和评估结果。